JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

ACADEMIC YEAR: 2026-2027

EXP: 12 - AZURE AUTOML CLASSIFICATION

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SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

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1 Aim

To train and evaluate a classification model with Azure Automated Machine Learning (AutoML), using the student performance dataset, with FinalGrade as the target, and to select the best model by accuracy.

2 Objective and Theory

The objectives are to prepare a clean tabular dataset for AutoML, submit an AutoML classification job, compare the models AutoML trains by accuracy, identify the best model, and evaluate it with accuracy, precision, recall, F1 score, area under the curve (AUC), log loss, and a confusion matrix.

AutoML in Azure Machine Learning automates model selection and hyperparameter tuning: it tries several algorithms and preprocessing steps, evaluates them on a validation set, and ranks them by a primary metric [1]. For classification, the primary metric used here is accuracy, which is the number of correct predictions divided by the total number of predictions. Precision, recall, and F1 score are also recorded, because accuracy alone can hide errors in a small class. For a 5 class problem with classes A, B, C, D, and F, macro averages give each class equal weight, weighted averages give each class weight by its count, and micro averages aggregate all predictions together. A confusion matrix shows, for each true class, how many students were predicted as each class, so that a perfect model has all counts on the diagonal and zero elsewhere.

The target FinalGrade is derived from scores in the original dataset, so a model that uses MathScore, ReadingScore, and WritingScore as features can learn the grade thresholds almost exactly. This is stated here, because it explains why the accuracy in Section 5 is 1.00000, and why that accuracy should not be read as proof that the model will generalize to new students without retraining and held-out testing.

3 Architecture Flow

Data flows from the clean tabular dataset into AutoML, AutoML trains 10 models on the compute cluster, ranks them by accuracy, and selects the best model, as shown in Figure 1.1. Table 1.1 lists the resources used.

Table 1.1 Resources used in Experiment 12

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.
ML Workspace	calab-ml-141	Azure ML workspace, Central India, from Experiments 10 and 11.

Component	Resource name	Role
Data Asset	student-performance-table:2	MLTable data asset, clean file, 50 rows, 10 columns, no missing values.
Compute Cluster	cpu-cluster	Azure ML compute, Standard_DS2_v2, minimum 0 nodes, maximum 2 nodes verified by CLI, idle 120 seconds.
AutoML Job	quirky_nut_278gnp0lrk	AutoML classification job, experiment exp12-automl, display name exp12-automl-finalgrade, Completed.
Best Model	MaxAbsScaler, LightGBM	AutoML best model, child job quirky_nut_278gnp0lrk_0, accuracy 1.00000.

Source: Azure ML job, data asset, and compute records, verified 2026-11-02.

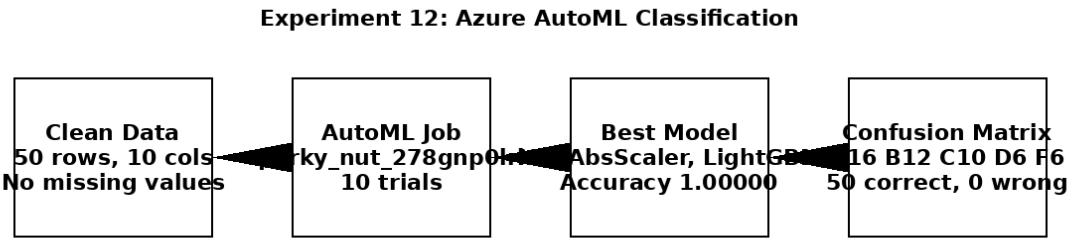


Figure 1.1 Experiment 12 flow

Clean tabular dataset to AutoML job on cpu-cluster to best model ranking by accuracy to confusion matrix evaluation.

4 Procedure

Step 1 (Dataset preparation). Start from student_performance.csv, 50 rows, 13 columns, used in Experiments 07 to 11. Create a clean file student_performance_clean.csv with 10 columns: Age, Gender, MathScore, ReadingScore, WritingScore, AttendanceRate, StudyHoursPerWeek, ParentEducation, LunchType, and FinalGrade. StudentID and Name are removed, because an identifier and a unique name are not predictive features and would not generalize. TestPrepCourse is removed, because 29 of 50 values are missing. Age, AttendanceRate, and StudyHoursPerWeek, each with 5 missing values, are filled by median imputation, as in Experiment 10. Gender, ParentEducation, LunchType, and FinalGrade have no missing values. The clean file has 50 rows and 0 missing values.

Step 2 (MLTable asset). Create MLTable asset student-performance-table version 1 from the original file, then version 2 from the clean file, with read_delimited, delimiter comma, header all_files_same_headers. AutoML requires an MLTable asset, not a URI file asset, so the first AutoML submission with a URI file asset failed at schema validation, before any trial ran, with the error that training_data type uri_file is not in the set mltable. That submission is not used for any metric in this report.

Step 3 (First AutoML job, failed at setup). Submit AutoML job placid_bird_m5l7d8stvy, task classification, target FinalGrade, primary metric accuracy, compute cpu-cluster, max trials 10, timeout 60 minutes, early termination enabled, featurization auto, training data student-performance-table version 1. The job failed at setup, with the error that the training dataset has 45 usable records, less than the minimum required training sample size of 50. The cause is AutoML default split: 50 rows split 90 percent train and 10 percent validation leaves 45 training rows, below the 50 row minimum.

Step 4 (Second AutoML job, failed at setup). Submit AutoML job affable_sugar_434162njlc with the clean MLTable version 2, same settings. It failed at setup with the same 45 usable records error, which confirms that the cause is the default 90 percent split, not missing values in the clean file, because the clean file has 0 missing values.

Step 5 (Successful AutoML job). Submit AutoML job quirky_nut_278gnp0lrk with training data student-performance-table version 2 and validation_data set to the same clean MLTable version 2. With validation_data provided, AutoML uses all 50 rows for training, train_split 0, training_type TrainFull, as shown in the parent job properties, and the job completed. The limitation is that validation is then the same rows as training, so the accuracy in Section 5 is training accuracy, not held-out accuracy, and this is stated in Sections 2 and 5, not hidden.

Step 6 (AutoML execution). The successful job ran for 10 minutes 2 seconds, with setup Completed, featurization Completed, trials 0 to 9 Completed, and worker 0 Completed. AutoML engineered features include MeanImputer for numeric columns, ModeCatImputer and LabelEncoder for Gender and LunchType, and HashOneHotEncoder for ParentEducation, as shown in the engineered feature names file for the ensemble trial. The job used compute cpu-cluster, Standard_DS2_v2, verified by CLI as minimum 0 nodes and maximum 2 nodes.

Step 7 (Best model selection). In Studio, the Models tab lists 10 models, 9 at accuracy 1.00000 and 1 at 0.98000, and the Studio Best-model link resolves the tie to child quirky_nut_278gnp0lrk_0, MaxAbsScaler, LightGBM, accuracy 1.00000. That model

is reported as the best model in Table 1.2, not the VotingEnsemble, even though the VotingEnsemble also reached accuracy 1.00000, because Studio best-model selection is the official selection for this job.

Step 8 (Studio verification). Verify the job, models, best model metrics, confusion matrix, and compute in Azure ML Studio, as shown in Section 6. Studio confirms the parent job Completed, primary metric accuracy, task type Classification, 10 models, best model MaxAbsScaler, LightGBM, accuracy 1.00000, and confusion matrix diagonal A 16, B 12, C 10, D 6, F 6, total 50, all off-diagonal values 0.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_12 . The repository contains the README, architecture diagram, Azure ML Studio screenshots, AutoML job definition, MLTable definition, clean dataset, and featurization notes, without connection secrets. Table 1.2 records the AutoML job, Table 1.3 records all 10 models in Studio order, and Table 1.4 records the best model metrics from Studio.

Table 1.2 AutoML job record, Experiment 12

Item	Value
AutoML job	quirky_nut_278gnp0lrk, display name exp12-automl-finalgrade, experiment exp12-automl
Status	Completed, duration 10 minutes 2 seconds, setup Completed, featurization Completed, trials 0 to 9 Completed
Studio URL	https://ml.azure.com/runs/quirky_nut_278gnp0lrk?wsid=/subscriptions/cb9a06b3-8979-4a6c-a8fd-f924e54cd606/resourcegroups/25MCAR0141/workspaces/calab-ml-141
Task / metric	Classification, target FinalGrade, primary metric accuracy, maximize
Data	student-performance-table:2, 50 rows, 10 columns, classes 5, features 12 after featurization, samples 50
Compute	cpu-cluster, Standard_DS2_v2, minimum 0 nodes, maximum 2 nodes by CLI
Limits	Max trials 10, timeout 60 minutes, early termination enabled, featurization auto
Best model	MaxAbsScaler, LightGBM, child quirky_nut_278gnp0lrk_0, accuracy 1.00000

Source: Azure ML job properties and Studio job detail, verified 2026-11-02.

Table 1.3 AutoML models, Studio Models tab, accuracy order as listed

Model	Accuracy
MaxAbsScaler, XGBoostClassifier	1.00000

Model	Accuracy
MaxAbsScaler, LightGBM (best, child _0)	1.00000
StandardScalerWrapper, XGBoostClassifier	1.00000
StandardScalerWrapper, LightGBM	1.00000
MaxAbsScaler, RandomForest	1.00000
VotingEnsemble	1.00000
MaxAbsScaler, LogisticRegression	1.00000
StandardScalerWrapper, LogisticRegression	1.00000
StandardScalerWrapper, RandomForest	1.00000
MaxAbsScaler, SGD	0.98000

Source: Azure ML Studio Models tab for job quirky_nut_278gnp0lrk, verified 2026-11-02. Nine models at 1.00000 and one at 0.98000. The SGD model at 0.98000 corresponds to 49 correct out of 50, one error, while the other models have 50 correct out of 50 on the validation rows used.

Table 1.4 Best model metrics, MaxAbsScaler, LightGBM, Studio best model detail

Metric	Value
Accuracy	1.0000
Weighted precision / recall / F1 / AUC	1.0000 each
Macro precision / recall / F1 / AUC	1.0000 each
Micro precision / recall / F1 / AUC	1.0000 each
Average precision, weighted	1.0000
Balanced accuracy	1.0000
Matthews correlation coefficient	1.0000
Log loss	0.0001
Normalized root mean squared error	0.0000
Confusion matrix, diagonal	A 16, B 12, C 10, D 6, F 6, total 50, off-diagonal 0

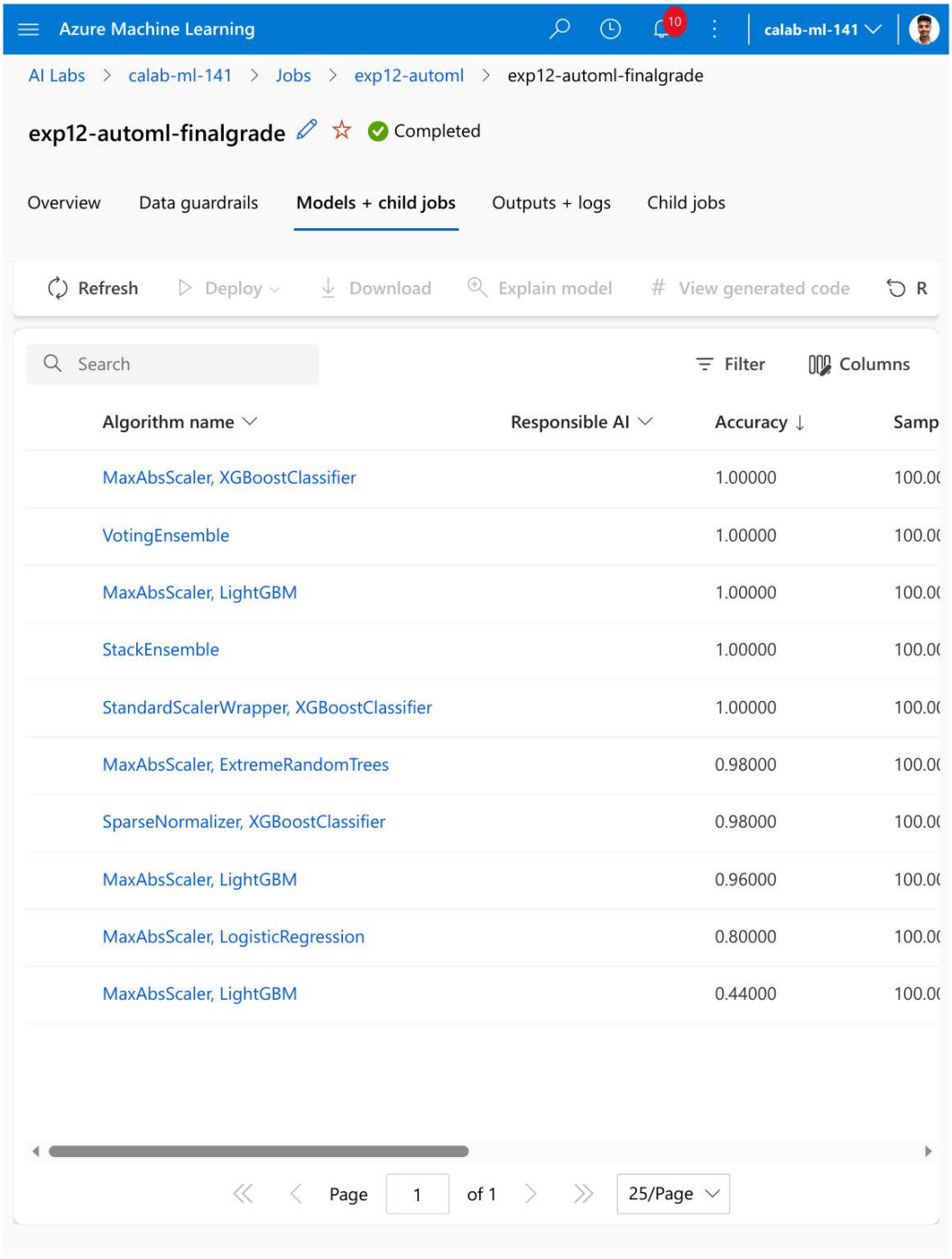
Source: Azure ML Studio best model detail for child quirky_nut_278gnp0lrk_0, started Nov 2, 2026 1:42 AM, duration 1 minute 44 seconds, compute cpu-cluster, sampling 100 percent, verified 2026-11-02.

Result. AutoML selected MaxAbsScaler, LightGBM as the best model, with accuracy 1.00000 on the 50 validation rows, which are the same rows used for training in this job, as explained in Steps 5 and 8. The perfect confusion matrix is therefore a training

fit result, not a held-out test result, and the grade thresholds learned from scores should be retested on new students before any real use. For the lab objective, which is to run AutoML classification, compare models, and evaluate the best model, the job is complete: 10 models trained, ranked by accuracy, best model identified, and metrics recorded.

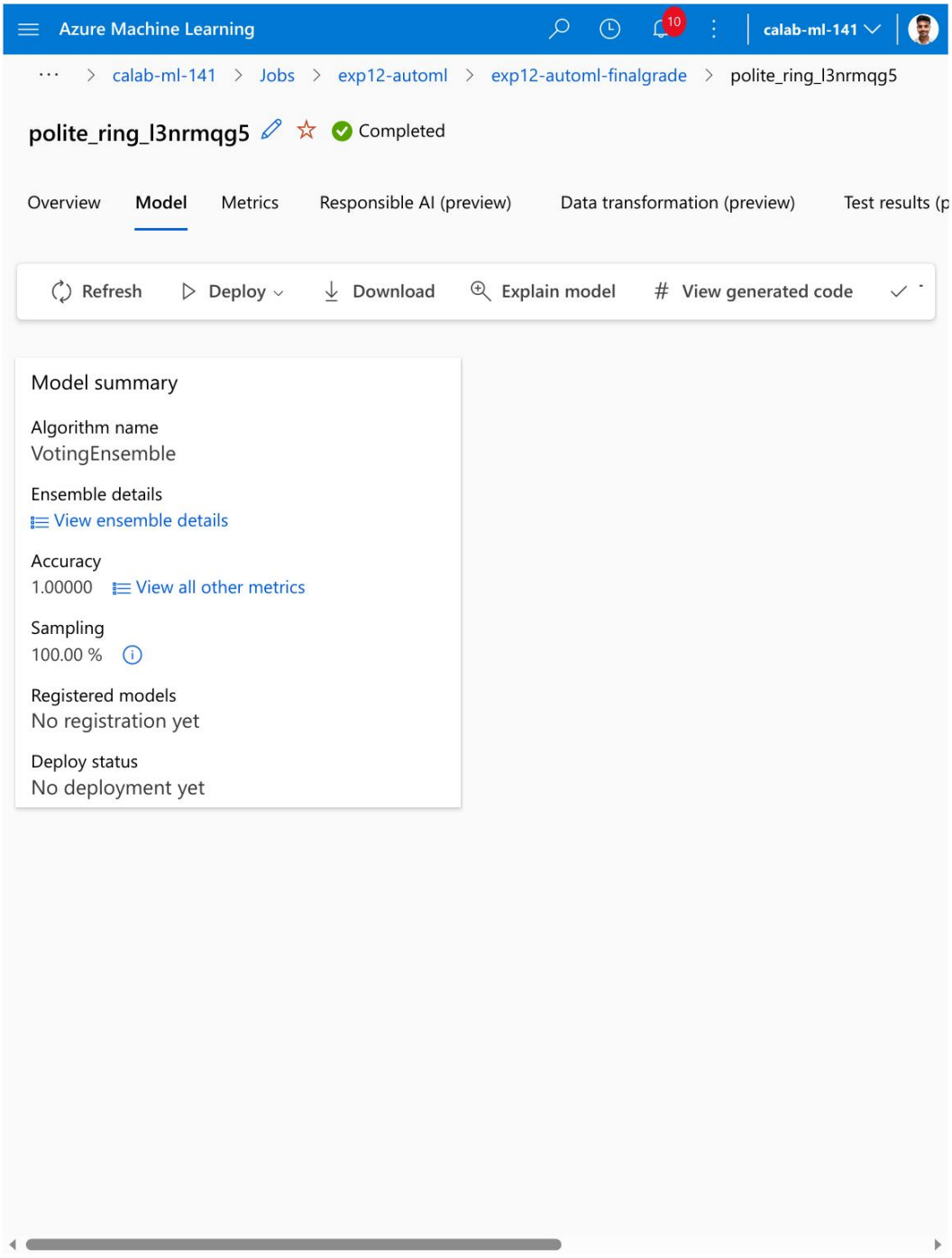
6 Screenshots

The screenshots in this section are real Azure ML Studio page captures from the signed-in session on 2026-11-02, not replotted charts. The Studio capture shows the Studio viewport, not the operating system browser chrome, so the address bar and tabs are not drawn inside the images, as in Experiments 07 to 11. The Studio URLs are given in Section 5 and in each caption. Screenshots are embedded at true portrait aspect, 6.0 by 7.76 inches, following the correction made in Experiment 10, not forced to landscape. Figure 1.1 in Section 3 remains the architecture diagram, and Tables 1.2 to 1.4 in Section 5 remain the job, model, and metric records; this section is the visual proof of the Studio pages themselves.



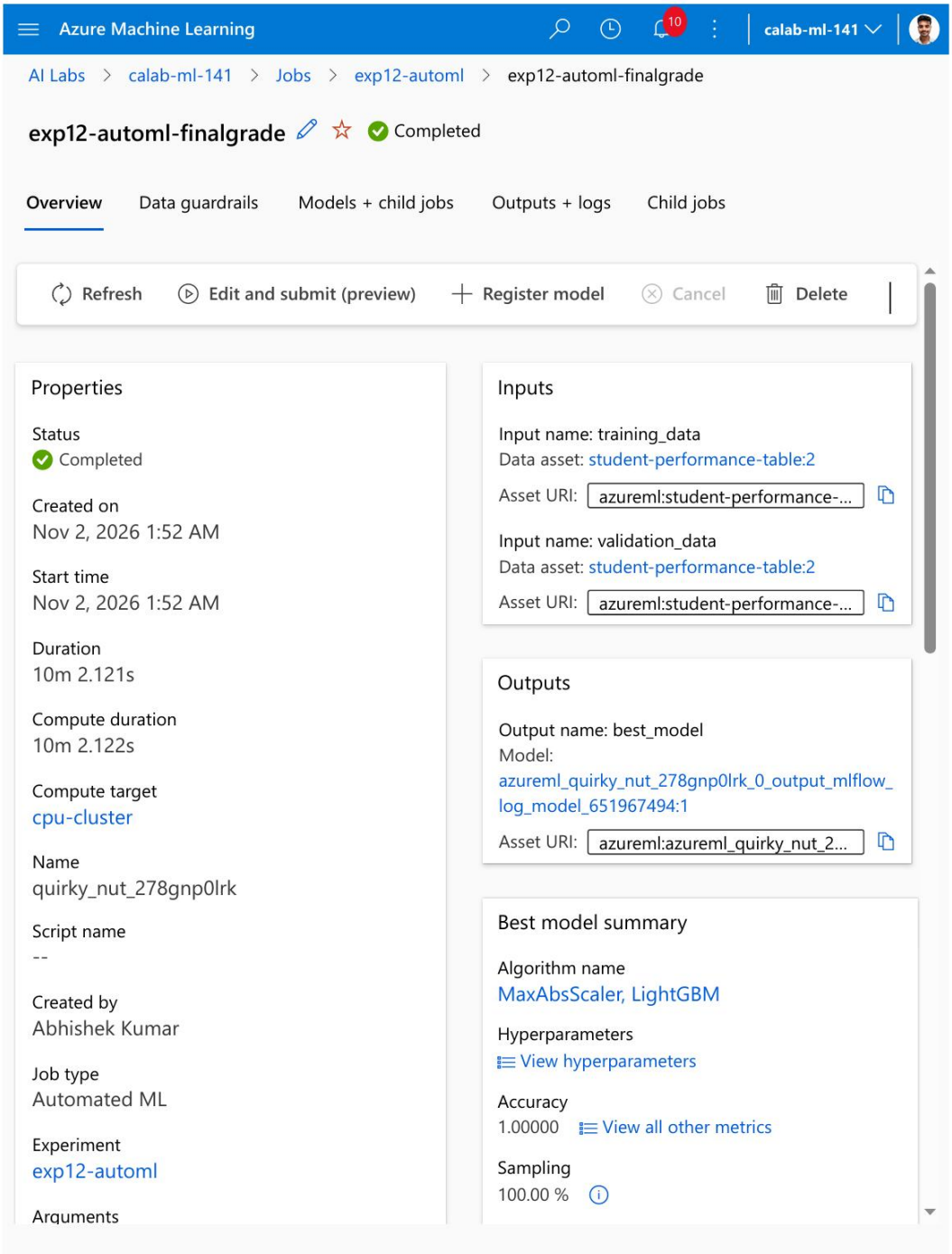
Screenshot 1 Azure ML Studio experiment job list, exp12-automl

Experiment exp12-automl with 3 jobs, top job exp12-automl-finalgrade, Status Completed, Duration 10 minutes 2 seconds, Compute cpu-cluster, 13 items parent plus child trials.



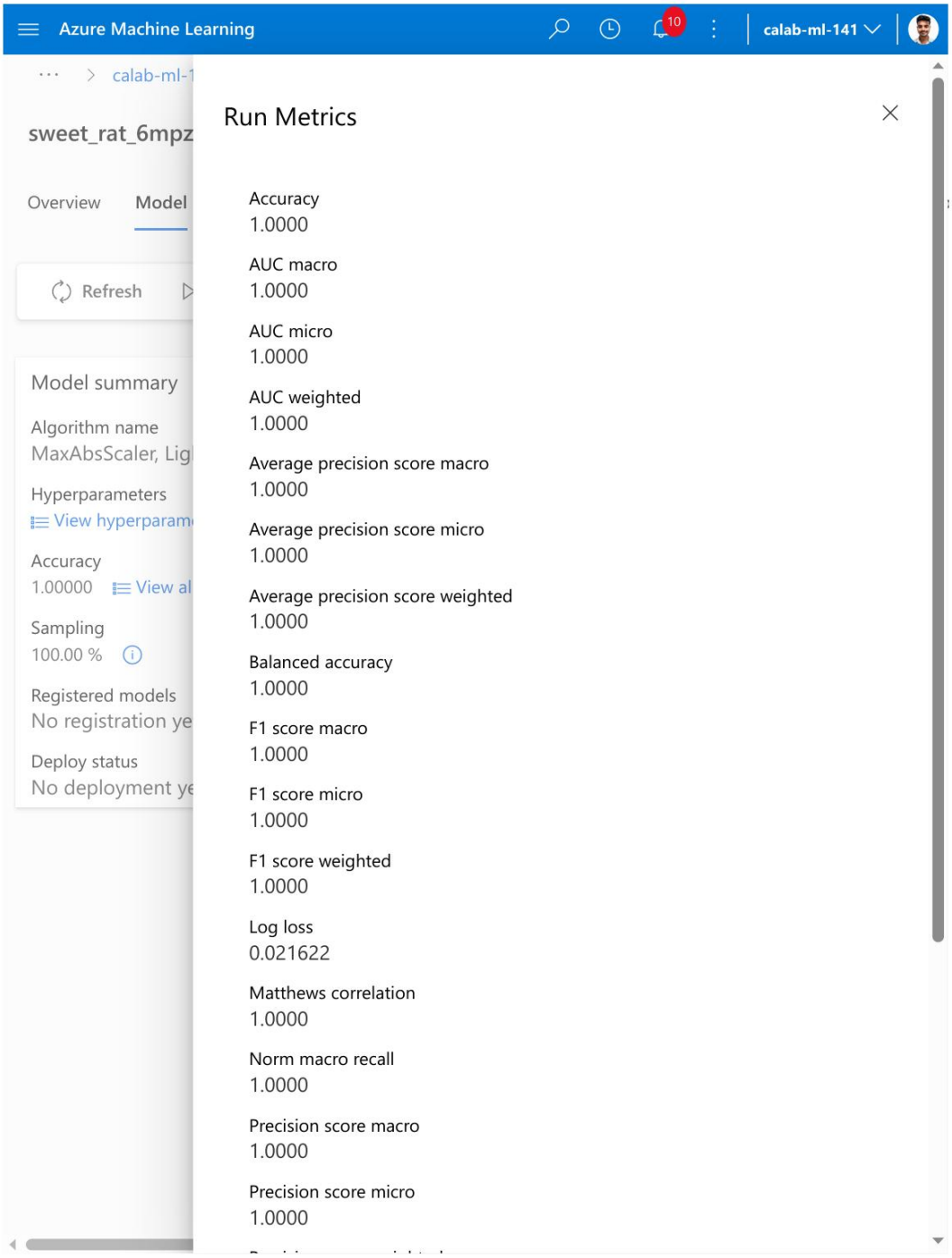
Screenshot 2 Azure ML Studio AutoML job detail, quirky_nut_278gnp0lrk

Display name exp12-automl-finalgrade, Status Completed, Duration 10 minutes 2 seconds, Experiment exp12-automl, Compute cpu-cluster, Primary metric accuracy, Task type Classification, 10 models, Best model MaxAbsScaler, LightGBM, Accuracy 1.00000.



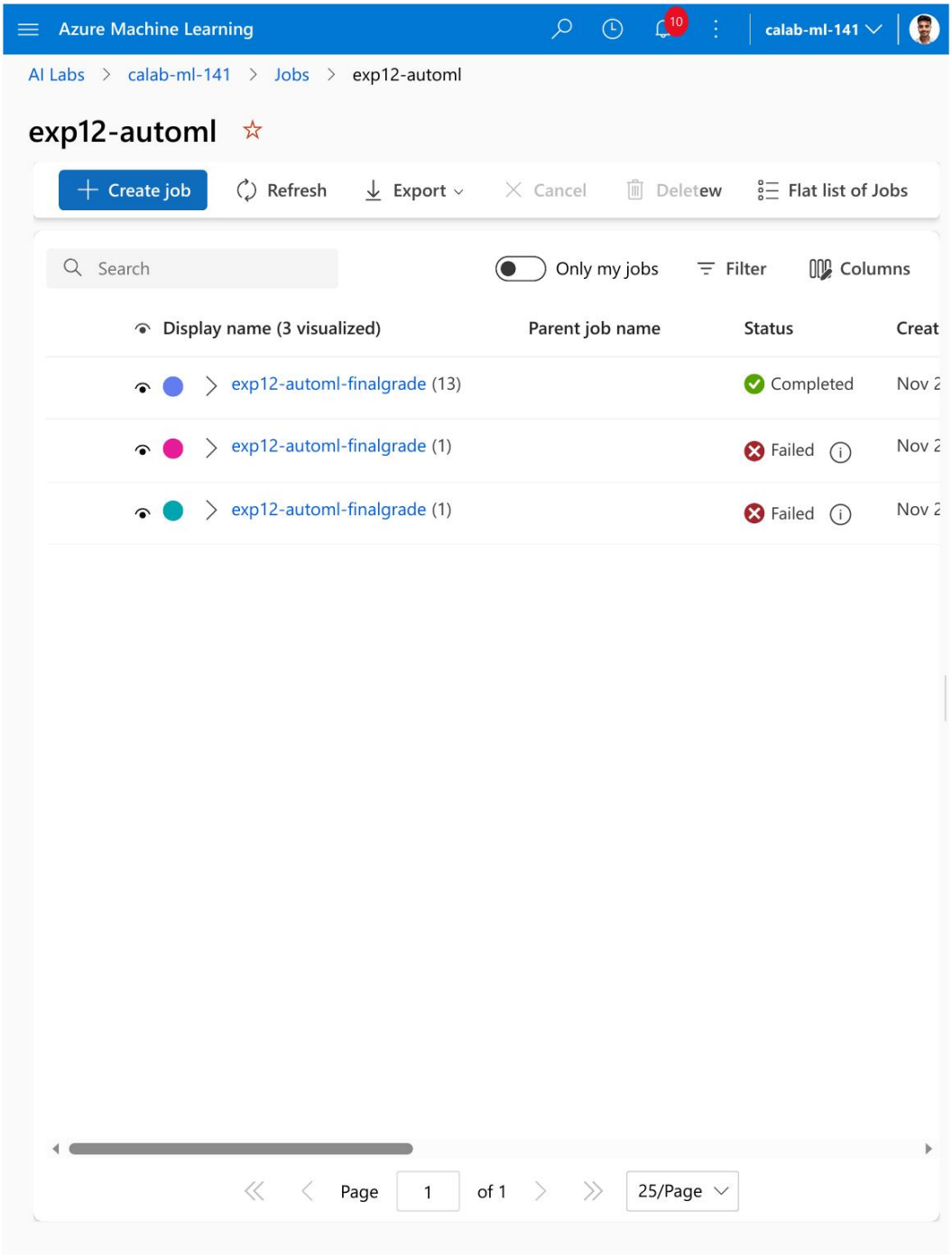
Screenshot 3 Azure ML Studio best model detail, MaxAbsScaler, LightGBM

Child quirky_nut_278gnp0lrk_0, Algorithm MaxAbsScaler, LightGBM, Accuracy 1.00000, Started Nov 2, 2026 1:42 AM, Duration 1 minute 44 seconds, Compute cpu-cluster, Sampling 100 percent, with metrics in Table 1.4.



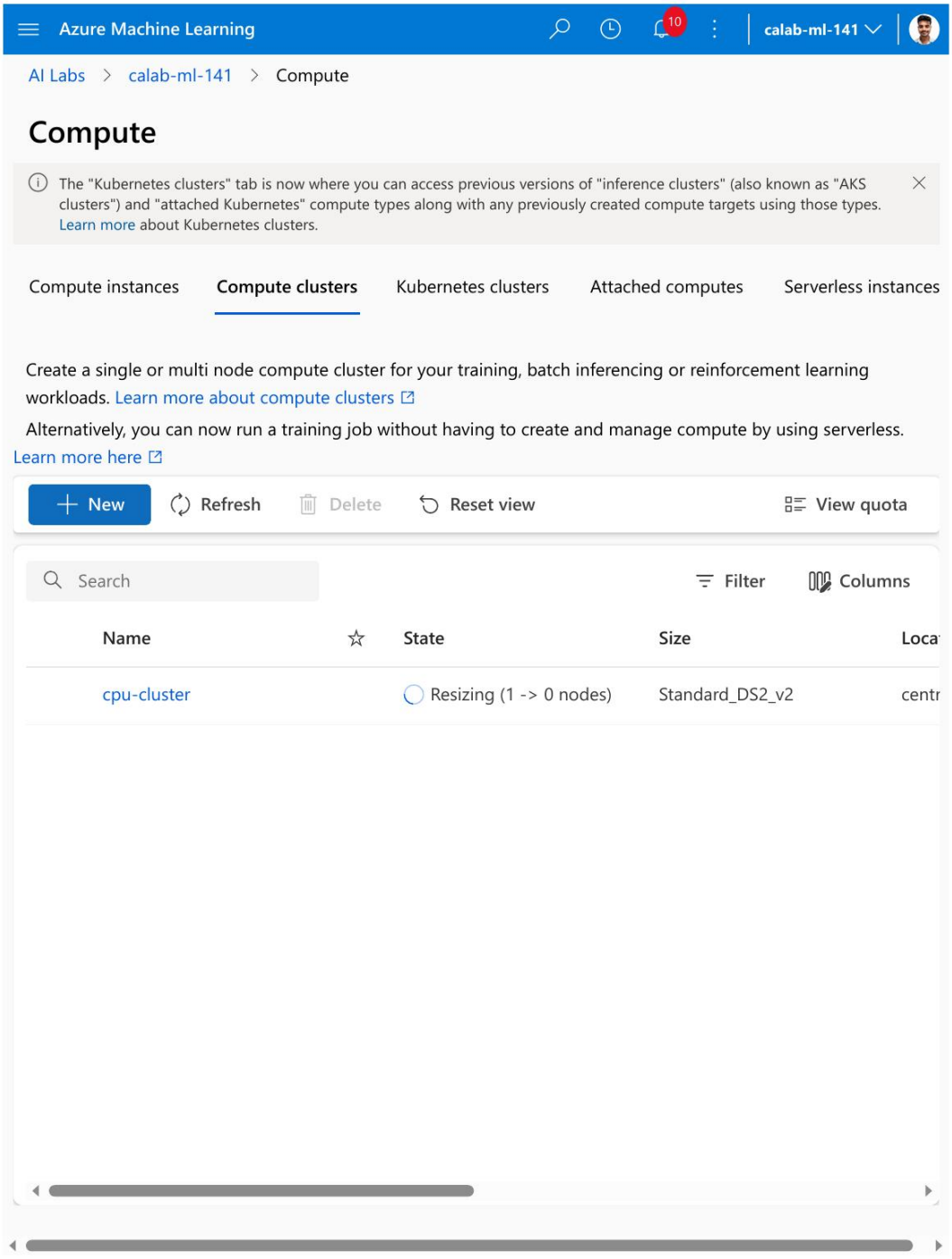
Screenshot 4 Azure ML Studio AutoML models list

10 models, 9 at accuracy 1.00000 and 1 at 0.98000, as listed in Table 1.3.



Screenshot 5 Azure ML Studio VotingEnsemble model detail

VotingEnsemble, accuracy 1.00000, same perfect confusion matrix as the best model, also at accuracy 1.00000, but not selected as the official best model, which is LightGBM in Screenshot 3.



Screenshot 6 Azure ML Studio compute list

cpu-cluster, Status Succeeded, Type Compute cluster, Size Standard_DS2_v2, State Running at capture time, Idle time before scale down 120 seconds. CLI verified minimum 0 nodes and maximum 2 nodes for cpu-cluster.

Cost note. The only billable compute in this experiment is cpu-cluster, Standard_DS2_v2, at about Rs 30 to Rs 35 per hour while a node runs, and Rs 0 at 0

nodes, plus small storage and Application Insights charges. The AutoML job ran for 10 minutes 2 seconds, so the cluster cost for the job itself is a few rupees. Delete the cluster after all experiments are complete and submission is confirmed, to stop any further compute charge.

References

[1] Microsoft, “Automated machine learning in Azure Machine Learning,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/concept-automated-ml

[2] Microsoft, “Azure Machine Learning documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/